MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, SILICON, DIODE ARRAY,
TYPE 1N6101
NON-TX, TX, AND TXV

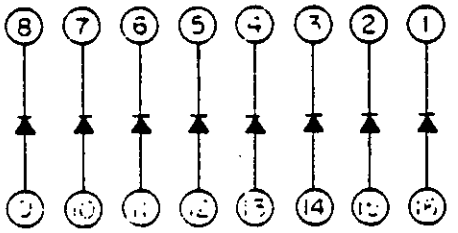
This specification is approved for use by the Rome Air Development Center, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for silicon, diode array. The prefix "TX" is used on devices submitted to and passing the special process-conditioning, testing, and screening as specified in 4.5.1 through 4.5.8. The prefix "TXV" is used on devices which have been subjected to and have passed the internal visual inspection specified in 4.6.

1.2 Physical dimensions. See figure 1.

1.3 Schematic configuration.



1N6101
ISOLATED EIGHT DIODE ARRAY

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center, RADC (RBRD), Griffiss AF Base, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Maximum ratings.

V_{BR} 1/ 2/	I_O 1/ 3/ T_A = 25°C	i_f (surge) 1/ t_p = 1.0S	P_T 4/ 5/ T_A = 25°C	T_{OP}	T_{stg}
75 Vdc	200 mAdc	1.0 A	600 mW	-65°C to +175°C	-65°C to +200°C

1/ Each diode.

2/ Pulsed: PW = 100 us maximum; duty cycle ≤ 20%.

3/ Derate at 1.33 mA/°C above 25°C.

4/ Derate at 2.67 mW/°C above 25°C.

5/ Maximum power per junction ≤ 400 mW.

1.5 Electrical characteristics, each diode.

Limits	V_F I_F = 100 mAdc	I_R V_R = 20 Vdc	C	t_{fr} I_F = 100 mAdc	t_{rr} $I_F = I_R$ = 10 mAdc R_L = 100 Ω i_{rr} = 1 mA
Min	—	—	—	—	—
Max	1.0 Vdc	25 nAdc	3 pF	15 ns	5 ns

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

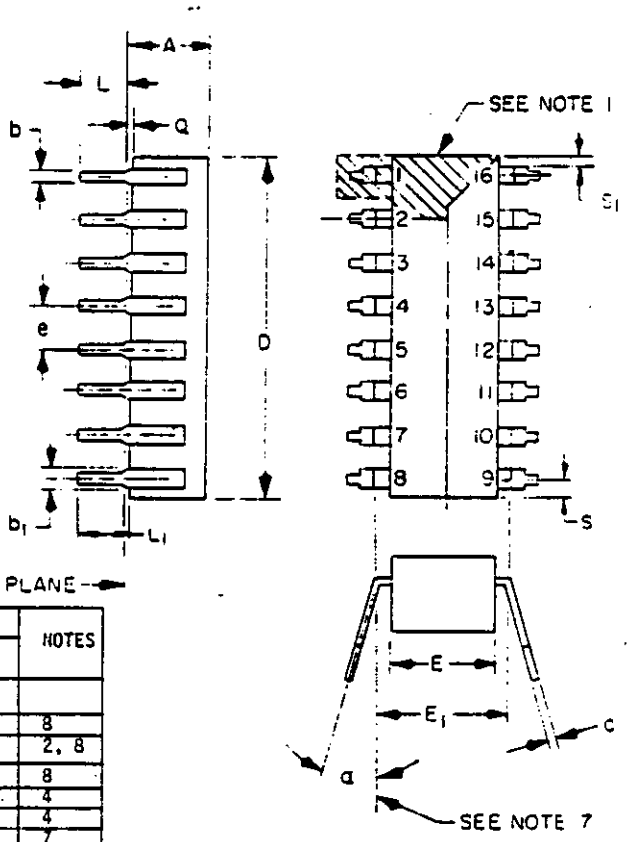
STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-750 - Test Methods for Semiconductor Devices.
MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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SYMBOL	INCHES		MILLIMETERS		NOTES
	MIN	MAX	MIN	MAX	
A		.200		5.08	
b	.014	.023	.36	.58	8
b_1	.030	.070	.76	1.78	2, 8
c	.008	.015	.20	.38	8
D		.785		19.94	4
E	.220	.310	5.59	7.87	4
E_1	.290	.320	7.37	8.13	7
e	.100 BSC		2.54 BSC		5, 9
L	.100	.200	2.54	5.08	
L_1	.150		3.81		
Q	.015	.060	.38	1.52	3
S		.098		2.49	6
S_1	.005		.13		6
α	0°	15°	0°	15°	

NOTES:
1. Index area; a notch or a pin one identification mark shall be located adjacent to pin one and shall be located within the shaded area shown. The manufacturer's identification shall not be used as a pin one identification mark.
2. The minimum limit for dimension b_1 may be .020 (.51 mm) for leads number 1, 8, 9 and 16 only.
3. Dimension Q shall be measured from the seating plane to the base plane.
4. This dimension allows for off-center lid, meniscus and glass overrun.
5. The basic pin spacing is .100 (2.54 mm) between centerlines. Each pin centerline shall be located within ±.010 (.25 mm) of its exact longitudinal position relative to pins 1 and 16.
6. Applies to all four corners (leads number 1, 8, 9 and 16).
7. Lead center when α is 0. E_1 shall be measured at the centerline of the leads.
8. All leads - Increase maximum limit by .003 (.08 mm) measured at the center of the flat, when lead finish A is applied.
9. Fourteen spaces.

FIGURE 1. Physical dimensions of diode, type 1N6101 (non-TX, TX, and TXV).

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3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. Diode array shall be of the design, construction, and physical dimensions shown on figure 1 and the case shall be hermetically sealed.

3.3.1 Lead material and finish. Lead material shall be Kovar or F-15 alloy and final finish shall be gold-plated. (Leads may be tin-coated if specified in the contract, and this requirement shall not be construed as adversely affecting the qualified-product status of the device or applicable "JAN" marking (see 6.2)).

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I, II, and III, and as follows:

3.4.1 Process-conditioning, testing, and screening for "TX" types. Process-conditioning, testing, and screening for the "TX" types shall be as specified in 4.5.

3.4.2 Internal visual (Precap) inspection and process-conditioning, testing, and screening of "TXV". The "TXV" device shall, in addition to all performance requirements, be internally visually inspected and process-conditioned, tested and screened in accordance with 4.6.

3.5 Marking. The following marking specified in MIL-S-19500 may be omitted from the body of the diode array at the option of the manufacturer:

a. Country of origin.
b. Manufacturer's identification.

3.5.1 "TX" marking. Devices in accordance with the "TX" requirements shall include the marking "JAN TX" preceding the type designation.

3.5.2 "TXV" marking. Devices in accordance with the "TXV" requirements shall be marked with "TXV" immediately following the "JAN" prefix.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of groups A, B, and C inspections. When specified in the contract, one copy of the quality conformance inspection data, pertinent to the device inspection lot, shall be supplied with each shipment by the device manufacturer (see 6.2).

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.3.3 Group C inspection. Group C inspection shall consist of the examinations and tests specified in table III and shall be performed on every tenth lot or every 6 months, whichever comes first.

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TABLE I. Group A inspection.

Examination or test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non TX	TX TXV		Min	Max	
Subgroup 1			7	7				
Visual and mechanical examination	2071				---	---	---	---
Subgroup 2			5	3				
Breakdown voltage	4021	I_R = 5 µA; PW = 100 us duty cycle ≤20%			V_{BR}	75	---	Vdc
Reverse current	4016	DC method; V_R = 20 Vdc			I_R		25	nAdc
Forward voltage	4011	PW = 300 us Duty cycle ≤2% (see 4.4.4) I_F = 100 mAdc			V_F	---	1.0	Vdc
Peak forward voltage	4011	PW = 150 ns Duty cycle ≤2% tr ≤10 ns (see 4.4.5) I_F = 100 mA			V_F	---	5.0	$V_{(PK)}$
Subgroup 3			10	7				
Capacitance	4001	V_R = 0 V; f = 1 MHz (each diode)			C	---	3	pF
Forward recovery time	4026	I_F = 100 mA R_S = 50 Ω, V_{fr} = 1.1 V tr ≤ 10 ns			t_{fr}	---	15	ns
Reverse recovery time	4031	Condition B I_F = I_R = 10 mA R_L = 100 Ω, i_{rr} = 1 mA			t_{rr}	---	5	ns
Forward voltage match	4011	I_F = 10 mA			V_F	---	10	mV
Subgroup 4			10	7				
High temperature operation:		T_A = 150°C			---	---	---	---
Reverse current	4016	DC method; V_R = 20 Vdc			I_R	---	50	µAdc
Low temperature operation:		T_A = -55°C						
Reverse current	4016	DC method; V_R = 37.5 Vdc			I_R		25	nAdc

TABLE II. Group B inspection.

Examination or test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non TX	TX TXV		Min	Max	
Subgroup 1			20	20				
Physical dimensions	2066	See figure 1			---	---	---	---
Subgroup 2			15	10				
Solderability	2026				---	---	---	---
Thermal shock (temperature cycling)	1051	Test condition C, 10 cycles; 15 minutes at extremes			---	---	---	---
Thermal shock (glass strain)	1056	Test condition A			---	---	---	---
Seal (leak-rate)	1071	Test condition G or H optional for fine leaks; test condition A, C-1, E, or F for gross leaks			---	---	$5x10^{-8}$	atm cc/s
Moisture resistance	1021	Omit initial conditioning			---	---	---	---

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TABLE II. Group B inspection - Continued.

Examination or test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non TX	TX TXV		Min	Max	
Subgroup 2 - Continued			15	10				
End points:								
Reverse current	4016	DC method; V_R = 20 Vdc			I_R	---	25	nAdc
Forward voltage	4011	I_F = 100 mAdc			V_F	---	1.0	Vdc
Subgroup 3			15	10				
Shock	2016	Nonoperating; 1,500 G, 0.5 ms, 5 blows in each orientation: X_1, Y_1, Y_2, and Z_1			---	---	---	---
Vibration, variable frequency	2056	20 G min, 100 to 2000 Hz			---	---	---	---
Constant acceleration	2006	20,000 G; in each orientation: X_1, Y_1, Y_2, and Z_1			---	---	---	---
End points: (Same as subgroup 2)								
Subgroup 4			20	20				
Terminal strength (lead fatigue)	2036	Test condition E, 3 oz weight; 3 bends of 45°						
Subgroup 5			20	20				
Surge current	4066	i_f (surge) = 1.0 A 10 surges at 1 per minute; t_p = 1 s cyclically choose four diodes from each array			---	---	---	---
End points: (Same as subgroup 2)								
Subgroup 6			7	5				
High-temperature life (nonoperating)	1032	T_{stg} = +200°C						
End points:								
Reverse current	4016	DC method; V_R = 20 Vdc			I_R	---	50	nAdc
Forward voltage	4011	I_F = 100 mAdc			V_F	---	1.1	Vdc
Subgroup 7			7	5				
Steady-state operation life	1027	DC blocking; V_R = 60 Vdc; T_A = 150°C; each diode (see 4.4.3)			---	---	---	---
End points: (Same as subgroup 6)								
Subgroup 8				15				
Bond strength test	---	Method 2011 of MIL-STD-883 Condition C or D (see 4.6.1)						

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TABLE III. Group C inspection.

Examination or test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non TX	TX TXV		Min	Max	
Subgroup 1			10	7				
Salt atmosphere (corrosion)	1041				---	---	---	---
Subgroup 2			10	10				
Resistance to solvents	---	Method 215 of MIL-STD-202 (see 4.4.2)			---	---	---	---
Subgroup 3			10	10				
Thermal shock (temperature cycling)	1051	Test condition C, 25 cycles; time at temperature extremes = 15 minutes, total test time = 72 hours max			---	---	---	---

4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I, II, and III, and as follows: (The electrical parameters and tests in this specification shall apply to each diode in an array and all diodes not under test shall be open circuited unless otherwise specified.)

4.4.1 Thermal shock (temperature cycling). Samples that have been subjected to the group B thermal shock test may be continued on test to 25 cycles in order to satisfy group C thermal shock test requirements. These samples shall be predesignated, and shall remain subjected to the group C 25-cycle acceptance evaluation after they have passed the group B 10-cycle acceptance criteria. The cumulative total of failures found during the 10-cycle test and during the subsequent interval up to 25 cycles shall be computed for 25-cycle acceptance criteria.

4.4.2 Resistance to solvents. The devices shall be tested in accordance with method 215 of MIL-STD-202 and as follows:

a. The brush shall be immersed with the device in the same solution and for the same length of time.
b. All areas of the device body where marking has been applied shall be brushed.
c. After completion of the test, there shall be no evidence of mechanical damage to the device, and markings shall have remained legible.

4.4.3 Steady-state operation life. The steady-state operation life test shall be performed as specified in test method 1027 of MIL-STD-750 and as follows: The junction shall be reverse biased at V_R = 60 Vdc. The ambient test temperature shall be T_A = 150°C. At the end of the high-temperature test time the ambient temperature shall be lowered. The test voltage shall be maintained on the devices until T_C = 30° ± 5°C is attained. After room ambient temperature has been established, the bias voltage shall be removed. After removal of the bias voltage, no other voltage shall be applied to the device before taking the specified end point measurements. The reverse current portion of the end point measurements shall be taken first.

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4.4.4 Forward voltage. This parameter shall be measured 90 microseconds after the leading edge of the pulse.

4.4.5 Peak forward voltage. During this test the maximum shunt capacitance across the diode shall be 19 picofarads and the equipment bandwidth shall be 80 megahertz minimum.

4.5 Process-conditioning, testing, and screening for "TX" types. The procedure for process-conditioning, testing, and screening the "TX" types shall be in accordance with MIL-S-19500 and 4.5.1 through 4.5.8.

4.5.1 High-temperature storage. All devices shall be stored for at least 48 hours at a minimum temperature (T_A) of 200°C.

4.5.2 Thermal shock (temperature cycling). All devices shall be tested in accordance with method 1051 of MIL-STD-750, test condition C, except that cycling duration shall be 10 continuous cycles and exposure time at temperature extremes shall be 15 minutes (minimum).

4.5.3 Acceleration. All devices shall be subjected to acceleration test in accordance with MIL-STD-750, method 2006, with the following exceptions: The test shall be performed one time in the Y_1 orientation only at a peak level of 20,000 G minimum. The 1-minute hold-time requirement shall not apply.

4.5.4 Hermetic seal tests. All devices shall be subjected to hermetic seal tests (fine leak followed by gross leak) with test conditions as specified in 4.5.4.1 and 4.5.4.2. Failed devices from either test shall be removed from the lot.

4.5.4.1 Fine-leak test. All devices shall be fine-leak tested in accordance with MIL-STD-750, method 1071, test condition G or H; except if condition H is used, the leak-rate rejection shall be 5×10^{-8} cubic centimeters of helium per second when measured at a differential pressure of one atmosphere.

4.5.4.2 Gross-leak test. All devices shall be tested for gross leaks in accordance with MIL-STD-750, method 1071, test condition A, C-1, E or F, except that the solution may be any suitable noncorrosive liquid at a minimum temperature of 90°C.

4.5.5 Preburn-in tests. The parameters V_F and I_R of table IV shall be measured and the data recorded for all devices in the lot. All devices shall be handled or identified such that the delta end points can be determined after the burn-in test. All devices which fail to meet these requirements shall be removed from the lot and the quantity removed shall be noted on the lot history.

TABLE IV. Burn-in test measurements (each diode in array).

Examination or test	MIL-STD-750		Symbol	Limits		Unit
	Method	Details		Min	Max	
Reverse current	4016	DC method; V_R = 20 Vdc	I_R	---	25	nAdc
Forward voltage	4011	I_F = 100 mAdc	V_F	---	1.0	Vdc

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4.5.6 Burn-in test. Each diode in the array shall be operated for 96 hours (minimum) under the following conditions:

$$V_R = 60\ Vdc,\ T_A = 150°C$$

4.5.7 Post burn-in test. The parameters I_R and V_F of table IV shall be retested in the specified order (within 96 hours) after burn-in and the data recorded for all devices in the lot. The specified reverse bias voltage shall be continuously maintained on the diode array during burn-in and cool-down to 25° ±3°C. After removal of reverse bias voltage, no other voltage shall be applied to the diode array before making the delta measurements. The parameters measured shall not have changed during the burn-in test from the initial value by more than the specified amount.

ΔI_R = 100 percent of initial reading or 10 nAdc, whichever is greater

ΔV_F = ±25 mVdc

4.5.8 Burn-in test failure (screening). All devices that exceed the delta (Δ) limit shall be removed from the lot and the quantity removed shall be noted on the lot history. If the quantity removed after burn-in test should exceed 10 percent of the total inspection lot on burn-in test, the entire lot shall be unacceptable for the "TX" type.

4.6 Internal visual (Precap) inspection and process-conditioning, testing, and screening of "TXV" types. The internal visual inspection shall be performed in accordance with method 2010, test condition B of MIL-STD-883 prior to encapsulation on a 100 percent basis and process-conditioning, testing, and screening shall be as specified in 4.5 through 4.5.8 and 4.6.1. The manufacturer shall permit the authorized government representative to witness concurrent with time of manufacturer's performance of these tests, the process-conditioning, testing, screening of the devices. Those conditioning and screening tests normally performed by a manufacturer as standard production tests, need not be repeated when these are predesignated and acceptable to the Government as being equal to or more severe than the test specified herein.

4.6.1 Bond strength test. The bond strength test shall be performed as specified in MIL-STD-883, method 2011, test condition C or D. The test sample may be selected randomly immediately following the internal visual inspection prior to sealing.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery. Preparation for delivery shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Ordering data. Procurement documents should specify the following:

a. Lead finish if other than gold plated (see 3.3.1).
b. Inspection data if required (see 4.3).

Custodian:
Air Force - 17

Review activities:
Air Force - 11, 99, 19, 85
DLA - ES

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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